

RUIYANG LIU

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🔗 RESEARCH INTERESTS

- **Neural rendering and reconstruction** - Digital human; NeRF (dynamic, editability); Implicit surface;
- **AI for Creativity** - AI content creation (CLIP, StyleGAN etc.) and interaction.

🎓 EDUCATION

ShanghaiTech University, Shanghai, China Sep. 2016 – Dec. 2022
MS. && Ph.D. in Computer Science (CS).

Dalian University of Technology, Dalian, China Sep. 2012 – July. 2016
B.S. in Software Engineering (SE).

👥 EXPERIENCE

Microsoft Research Asia May. 2015 – Jan. 2016
R&D Intern Topic: C++/CLI, .NET; Learning basics (Face detection/recognition).

Google Inc July 2019 – Dec. 2019 and July 2021 – Oct. 2021
SWE Intern Topic: Differentiable neural architecture search (2019); Compiler optimization (2021).

Runway ML May 2021 – Aug. 2021
Research Intern Topic: Interactive video stylization.

★**Pixel Lab (Tencent America)** May 2022 – Dec. 2022
Graphics Research Intern Topic: Cloud-native neural rendering engine design and implementation.

♡ PUBLICATION

SINCE: Stylizable Interactive Neural 4D Creation Engine Sep. 2021 – Apr. 2022
Ruiyang Liu, Huangjie Yu, Anpei Chen, Jingyi Yu (TOG under review)
Keywords: neural 4D representation, interactive stylization, few-shot

SofGAN: A Portrait Image Generator with Dynamic Styling Dec. 2019 – Sep. 2020
Anpei Chen, Ruiyang Liu*, Ling Xie, Hao Su, Jingyi Yu (TOG 2021)*
Keywords: image generation, GAN, implicit representation, semantic analysis

Syntropic Counterpoints: Metaphysics of The Machine Dec. 2020 – Feb. 2021
Ruiyang Liu, Shengcheng Luo, Predrag. K. Nikolic (ACM MM 2021)
Keywords: multimodal user interface, human-AI cocreation

Learning to Dodge A Bullet Dec. 2017 – Mar. 2018
Shi Jin, Ruiyang Liu, Jinwei Ye, Yu Ji, Jingyi Yu (ECCV 2018)
Keywords: bullet-time effects, view morphing, semantic analysis

Automatic 3D Indoor Scene Modeling from Single Panorama Dec. 2016 – Nov. 2017
Yang Yang, Shi Jin, Ruiyang Liu, Singbing Kang, Jingyi Yu (CVPR 2018)
Keywords: panoramic 3D reconstruction, semantic analysis

⚙️ SKILLS

- **Programming:** ML (PyTorch, Tensorflow) > Graphics (OpenGL == **CUDA** > **mantaflow** > taichi) > VR/AR (Unreal, **ARKit**) > Web (JS, **WebGL**, ThreeJS, p5)
- **Misc:** Screenwriting, Photography (PS + AI + LR), 3D Modeling (Blender, 3DS Max)

⚙️ PROJECTS

Multimodal AI Companion for Interactive Fairytale Co-creation Dec 2020 - April 2021

Interactive AI system **Technique:** Few-shot text generation; Scene Graph; SketchRNN; P5;

This project investigates the potentials of using AI as a creative medium, and adopt techniques including dialogue generation and transformation between text, audio and visual contents.

★ Portrait generator with dynamic stylizing Dec 2019 - July 2020

3D vision **Technique:** Implicit Scene Representation; StyleGANs.

In this work, we combine implicit geometry representation, semantic representation and GAN based image generator to a new portrait generation framework that enables high quality portrait image generation with independently controllable geometry and texture attributes. Research paper accepted by **TOG (2021)**.

3D Face Geometry from a Single Side Shot July 2019 - Nov 2019

3D vision **Technique:** 3D reconstruction; Geometry images; StyleGANs.

In this work, we adopt StyleGAN, MVS and geometry image to generate for high-resolution 3D face generation.

Probabilistic SfM with objects and pose May 2019

3D vision **Technique:** Object discovery; Probabilistic Model; SfM.

Based on PSfMO(Probabilistic SfM with Objects), we fuse learning based 6D object pose estimation to better estimate camera pose while the DLT method failed to come to a unique solution.

Semantic Re-Disparity redistribution for light field stereo Dec. 2018 - April 2019

Computational Photography **Technique:** MRF; Light-field rendering; Probabilistic modeling.

We adopt scene semantic labeling for finding an appropriate disparity label set that accelerates matching and improves accuracy.

★ Light-field Capture Systems Aug. 2016 – Nov. 2017

Software Development **Technique:** WPF, OpenFrameworks, WebGL, ThreeJS

Real-time light-field capture system with 70+ cameras for both static picture and video.

★ Physical based simulator March 2017

Animation, PBR **Technique:** Graphics; Animation

Design and develop a physical based simulator for realistic animation and rendering, including **rigid body dynamics**, **cloth simulation** and **fluid simulation**.

Time-lapse Relighting Dec. 2016

Graphics **Technique:** Graphics; Computational animation

Propose a relighting pipeline based on photometric stereo and Time-lapse Illumination Decomposition. Analysis sunlight, sky-light component and scene geometry from time-lapse videos and blend a new object into the scene with realistic illumination effects.

Refine Fish-eye Distortion with GAN Dec. 2016

Vision **Technique:** Distortion correction; GAN

Follow the idea of Single Image Super Resolution(SISR) based on GAN [Ledig, Christian, et al 2016], we apply GAN architecture to undistortion task. Improve geometry accuracy and image quality simultaneously with coarse image rectified with estimated camera parameter.